TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Focal Length of a Convex Lens by the Displacement Method

Object and screen stay fixed; the lens alone moves between its two image-forming positions — their separation d and the fixed span D give f = (D² − d²)/4D, free of bench index error.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 11 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to find f from the two lens positions that image a fixed object onto a fixed screen. Fix object and screen a distance D apart (D > 4f). By conjugacy (Experiment 10) there are exactly **two** lens positions between them that give a sharp image — one nearer the object (magnified image), one nearer the screen (diminished image) — and the two positions are mirror images: the u of the first is the v of the second. If the lens travels a distance d between the positions, algebra on the lens formula gives

$$f = (D^2 - d^2) / (4D)$$

The method's beauty is what it does *not* need: no absolute distance from the lens's optical centre is ever measured (both D and d are differences between scale readings), so the bench index error that haunts Experiment 10 cancels out. A second fruit: the two image sizes I_1 and I_2 obey $I_1 I_2 = O^2$ (O = object size), so the object's size can be found from the images alone.

2. Apparatus and procedure

- **Apparatus:** optical bench, convex lens, illuminated object, screen — as Experiment 10.
- Set object and screen about 100 cm apart (D must exceed 4f; estimate f first by the distant-object method). Fix them; they do not move again.
- Slide the lens from the object end: find the first sharp image (magnified); record the lens position. Continue toward the screen: find the second sharp image (diminished); record the position. d is the difference.
- Measure both image sizes for the $O^2 = I_1 I_2$ check.
- Repeat for another D; compute f each time and average.

3. Worked calculation (model data)

RESULT

D = 100.0 cm (object-to-screen). Lens positions: 20.0 cm and 80.0 cm on the bench scale → d = 60.0 cm.

$f = (D^2 - d^2)/(4D) = (10000 - 3600)/400 = 6400/400 =$ **16.0 cm**.

Consistency: with f = 16 cm, the two settings are u = 20, v = 80 and u = 80, v = 20 (check: 20 × 80/100 = 16 ✓). Image sizes: I_1 = 4.0 cm (magnified, m = 4), I_2 = 0.25 cm (m = 1/4) → $O = \sqrt{(I_1 I_2)} = \sqrt{(1.0)} = 1.0$ cm, matching the object arrow — the method auditing itself.

4. Precautions

- D must be greater than 4f: at D = 4f the two positions merge into one (u = v = 2f), and below 4f no lens position images on the screen at all.
- Object and screen must not be disturbed between the two lens settings — the whole method rests on their fixity.
- Focus sharply at both positions by bracketing; the diminished image is small and dim, so screen the room light if needed.
- Read lens positions to the same index mark on its upright both times (d is a difference — the mark's absolute offset cancels, its consistency does not).
- Keep the lens upright and centred on the bench axis at both positions.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why are there exactly two lens positions, and how are their (u, v) pairs related?
2. With D = 90.0 cm and d = 30.0 cm, compute f, then find the two u values.
3. Why does the bench index error cancel in this method but not in Experiment 10?
4. What happens as D approaches 4f, and what does the single merged position correspond to?
5. The two images measure 6.0 cm and 1.5 cm. Find the object size and the two magnifications.
6. Show that $d^2 = D^2 - 4Df$ requires D ≥ 4f for a real d, and interpret the algebra physically.

Answer key

1. Because the lens equation with u + v = D fixed is quadratic in u: its two roots are the two positions. The roots are u and D − u, so the second position's object distance is the first position's image

distance — the conjugate swap, guaranteed by reversibility.

2. f = (8100 − 900)/(360) = 7200/360 = 20.0 cm. The positions split D as u and D − u with u(D − u) = fD = 1800 → $u^2 - 90u + 1800 = 0$ → u = (90 ± 30)/2 = 30 cm or 60 cm. (Note the roots differ by exactly d = 30 cm, as they must.)
3. Both D and d are differences of pairs of scale readings taken with the same uprights: any constant offset in where an index sits relative to the true plane appears in both readings and subtracts away. In Experiment 10, u and v are single intervals measured from a lens index whose offset from the optical centre enters directly.
4. The two positions approach each other and merge at the midpoint, where u = v = D/2 = 2f: the same-size image setting of Experiment 10. Below D = 4f there is no position at all.
5. O = $\sqrt{(I_1 I_2)}$ = $\sqrt{(6.0 \times 1.5)}$ = $\sqrt{9.0}$ = 3.0 cm. Magnifications: m_1 = 6.0/3.0 = 2.0 and m_2 = 1.5/3.0 = 0.50 — reciprocals, as conjugacy demands ($m_1 m_2 = 1$).
6. From $f = (D^2 - d^2)/4D$, $d^2 = D^2 - 4Df$: a real displacement needs $d^2 \geq 0$, i.e. $D \geq 4f$. Physically: object and screen must be far enough apart for two distinct conjugate settings to exist between them; at the boundary they coalesce into the single 2f–2f setting.
